

Computational biophysics

	Microcanonical $\Omega(E, V, N)$	Canonical $Z(T, V, N)$	Grand canonical $\mathcal{Z}(T, V, \mu)$
$\frac{S}{k}$	$\ln \Omega$	$\left(\frac{\partial(T \ln Z)}{\partial T}\right)_{V, N}$	$\left(\frac{\partial(T \ln \mathcal{Z})}{\partial T}\right)_{V, \mu}$
F	$E - kT \ln \Omega$	$-kT \ln Z$	$kT\mu^2 \left(\frac{\partial(\mu^{-1} \ln \mathcal{Z})}{\partial \mu}\right)_{T, V}$
U	Fixed ($=E$)	$kT^2 \left(\frac{\partial(\ln Z)}{\partial T}\right)_{V, N}$	$-\left(\frac{\partial(\ln \mathcal{Z})}{\partial \beta}\right)_{\beta \mu, V}$
N	Fixed	Fixed	$kT \left(\frac{\partial(\ln \mathcal{Z})}{\partial \mu}\right)_{T, V}$
kT	$\left(\frac{\partial(\ln \Omega)}{\partial E}\right)_{V, N}^{-1}$	Fixed	Fixed
$\frac{\mu}{kT}$	$-\left(\frac{\partial(\ln \Omega)}{\partial N}\right)_{E, V}$	$-\left(\frac{\partial(\ln Z)}{\partial N}\right)_{T, V}$	Fixed
P	$kT \left(\frac{\partial(\ln \Omega)}{\partial V}\right)_{E, N}$	$kT \left(\frac{\partial(\ln Z)}{\partial V}\right)_{T, N}$	$\frac{kT}{V} \ln \mathcal{Z}$
$\frac{C_V}{k}$	$-\beta^2 \left(\frac{\partial^2(\ln \Omega)}{\partial E^2}\right)_{V, N}^{-1}$	$\beta^2 \left(\frac{\partial^2(\ln Z)}{\partial \beta^2}\right)_{V, N}$	$T \left(\frac{\partial^2(T \ln \mathcal{Z})}{\partial T^2}\right)_{V, \mu}$
$(\Delta N)^2$	0	0	$\left(\frac{\partial^2(\ln \mathcal{Z})}{\partial(\beta \mu)^2}\right)_{\beta, V}$
$(\Delta E)^2$	0	$\left(\frac{\partial^2(\ln Z)}{\partial \beta^2}\right)_{V, N}$	$\left(\frac{\partial^2(\ln \mathcal{Z})}{\partial \beta^2}\right)_{\beta \mu, V}$

Protein's geometry

Centre of mass $\vec{R}_{cm} = \frac{\sum_{i=1}^N m_i \vec{r}_i}{\sum_{i=1}^N m_i}$

Radius of gyration $r_g = \sqrt{\frac{\sum_{i=1}^N m_i (\vec{r}_i - \vec{R}_{cm})^2}{\sum_{i=1}^N m_i}}$

$RMSD(t) = \sqrt{\frac{1}{N} \sum_{i=1}^N (\vec{r}_i(t) - \vec{r}_i(0))^2}$

$RMSF_i = \sqrt{\langle \Delta r_i^2 \rangle} = \sqrt{\frac{1}{M} \sum_{f=1}^M (\vec{r}_{i,f} - \langle \vec{r}_i \rangle)^2}$

$B_i = \frac{8\pi^2}{3} RMSF_i^2$

Semi-empirical force fields

Bond stretching

Harmonic $U(r_{AB}) = \frac{1}{2} k_{AB} (r_{AB} - r_{AB,eq})^2$

Anarmonic $U(r_{AB}) = \frac{1}{2} \left[k_{AB} + k_{AB}^{(3)} (r_{AB} - r_{AB,eq}) \right] (r_{AB} - r_{AB,eq})^2$

Quartic correction $U(r_{AB}) = \frac{1}{2} \left[k_{AB} + k_{AB}^{(3)} (r_{AB} - r_{AB,eq}) + k_{AB}^{(4)} (r_{AB} - r_{AB,eq})^2 \right] \cdot (r_{AB} - r_{AB,eq})^2$

Morse $U(r_{AB}) = D_{AB} \left[1 - e^{-\alpha_{AB}(r_{AB} - r_{AB,eq}^2)} \right]^2$

Valence angle bending

Potential $U(\theta_{ABC}) = \frac{1}{2} [k_{ABC} + k_{ABC}^{(3)} (\theta_{ABC} - \theta_{ABC,eq}) + k_{ABC}^{(4)} (\theta_{ABC} - \theta_{ABC,eq})^2 + \dots] (\theta_{ABC} - \theta_{ABC,eq})^2$

$U(\theta_{ABC}) = \sum_{\{j\}_{ABC}} k_{j,ABC}^{fourier} [1 + \cos(j\theta_{ABC} + \psi_j)]$

Fourier $k_{j,ABC}^{fourier} = \frac{2k_{ABC}^{harmonic}}{j^2}$

Torsions

Potential $U(\omega_{ABCD}) = \frac{1}{2} \sum_{\{j\}_{ABCD}} V_{j,ABCD} [1 + (-1)^{j+1} \cos(j\omega_{ABCD} + \psi_{j,ABCD})]$

Improper $U(\omega_{ABCD}) = \frac{1}{2} \sum_{\{j\}_{ABCD}} V_{j,ABCD} [1 + (-1)^{j+1} \cos(j\omega_{ABCD} + \psi_{j,ABCD})]$

Van der Waals

Lennard-Jones $U(r_{AB}) = 4\epsilon_{AB} \left[\left(\frac{\sigma_{AB}}{r_{AB}}\right)^{12} - \left(\frac{\sigma_{AB}}{r_{AB}}\right)^6 \right]$

Morse $U(r_{AB}) = D_{AB} \left[1 - e^{-\alpha_{AB}(r_{AB} - r_{AB,eq}^2)} \right]^2$

Hill $U(r_{AB}) = \epsilon \left[\frac{6}{\beta_{AB}-6} e^{\beta_{AB} \frac{1-r_{AB}}{r_{AB}}} - \frac{\beta_{AB}}{\beta_{AB}-6} \left(\frac{r_{AB}}{r_{AB}}\right)^6 \right]$

Electrostatic interactions

Distribution of charges $U_{AB} = \sum_A \sum_{B>A} \vec{M}^{(A)} \vec{V}^{(B)}$

Point like $U_{AB} = \frac{q_A q_B}{\epsilon_{AB} r_{AB}}$

Dipolar interactions $U_{AB/CD} = \frac{\mu_{AB} \mu_{CD}}{\epsilon_{AB/CD} r_{AB/CD}^3} (\cos \chi_{AB/CD} - 3 \cos \alpha_{AB} \cos \alpha_{CD})$

Parameterization

Parameters $Z = \sqrt{\frac{\sum_i^{observables\ occurrences} (calc_{i,j} - exp_{i,j})^2}{\sum_j w_i^2}}$

$\sigma_{AB} = \sigma_A + \sigma_B$ $\epsilon_{AB} = \sqrt{\epsilon_A \epsilon_B}$

Classical mechanics

Newton's laws

$\vec{F} = m\vec{a}$ $\vec{F}_{BA} = -\vec{F}_{AB}$

$\vec{v}(t) = \frac{d\vec{r}}{dt}$ $\vec{a}(t) = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$ $m \frac{d^2\vec{r}}{dt^2} = \vec{F}$

Force acting on atom $\vec{F}_i(\vec{r}_1, \dots, \vec{r}_N, \vec{r}_i) = \sum_{j \neq i} \vec{F}_{ij}(\vec{r}_i - \vec{r}_j) + \vec{F}^{(ext)}(\vec{r}_i, \dot{\vec{r}}_i)$

Bond stretching: $U = \frac{k_l}{2} (l - l^0)^2$

Bond bending: $U = \frac{k_\theta}{2} (\theta - \theta^0)^2$

Bond torsion: $U = k_\phi [1 + \cos(n\phi - \phi^0)]$

Van der Waals interactions: $U = \left[\frac{a_{ij}}{r_{ij}^{12}} - \frac{b_{ij}}{r_{ij}^6} \right]$

Electrostatic interactions: $U = \frac{332 q_i q_j}{\epsilon r_{ij}}$

$\vec{p}_i = m_i \dot{\vec{v}}_i = m \dot{\vec{r}}_i$ $\vec{F}_i = m_i \ddot{\vec{r}}_i = \dot{\vec{p}}_i$

$\vec{x}(t) = \{\vec{r}_1(t), \dots, \vec{r}_N(t), \vec{p}_1(t), \dots, \vec{p}_N(t)\}$

Lagrangian formulation

$\vec{F}_i(\vec{r}_1, \dots, \vec{r}_N) = -\Delta_i U(\vec{r}_1, \dots, \vec{r}_N)$

$W_{AB} = \int_A^B \vec{F}_i d\vec{l} = U_A - U_B = -\Delta U_{AB}$ $\oint \vec{F}_i d\vec{l} = 0$

Kinetic energy $K(\vec{r}_1, \dots, \vec{r}_N) = \frac{1}{2} \sum_i m_i \dot{\vec{r}}_i^2$

$\mathcal{L}(\vec{r}_1, \dots, \vec{r}_N, \dot{\vec{r}}_1, \dots, \dot{\vec{r}}_N) = K(\vec{r}_1, \dot{\vec{r}}_N) - U(\vec{r}_1, \dots, \vec{r}_N)$

Euler-Lagrange $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\vec{r}}_i} \right) - \frac{\partial \mathcal{L}}{\partial \vec{r}_i} = 0$

$E = \frac{1}{2} \sum_i m_i \dot{\vec{r}}_i^2 + U(\vec{r}_1, \dots, \vec{r}_N)$

$quad \frac{dE}{dt} = 0$

Generalized coordinates

$q_\alpha = f_\alpha(\vec{r}_1, \dots, \vec{r}_N)$ $\alpha = 1, \dots, 3N$ $\vec{r}_i = \vec{q}_i(q_1, d, \dots, q_{3N})$ $i = 1, \dots, N$

$\dot{\vec{r}}_i = \sum_{\alpha=1}^{3N} \frac{\partial \vec{r}_i}{\partial q_\alpha} \dot{q}_\alpha$ $\mathcal{L}(q, \dot{q}) = \frac{1}{2} \sum_{\alpha=1}^{3N} \sum_{\beta=1}^{3N} G_{\alpha\beta} \dot{q}_\alpha \dot{q}_\beta - U(q_1, \dots, q_{3N})$

Classical mechanics (contd)

Legendre transforms

$s = f'(x) \equiv g(x)$ $f'(x) = g(x) = s \Rightarrow x = g^{-1}(s)$

$b(g^{-1}(s)) = f(g^{-1}(s)) - sg^{-1}(s) \equiv \tilde{f}(s) = f(x(s)) - sx(s)$

$\tilde{f}(s_1, \dots, s_n) = f(x_1(s_1, \dots, s_n), \dots, x_n(s_1, \dots, s_n)) - \sum_i s_i x_i(s_1, \dots, s_n)$

Hamiltonian formulation

$\mathcal{H}(\vec{r}_1, \dots, \vec{r}_N, \vec{p}_1, \dots, \vec{p}_N) = -\tilde{\mathcal{L}}(\vec{r}_1, \dots, \vec{r}_N, \vec{p}_1, \dots, \vec{p}_N)$

$\mathcal{H}(\vec{r}_1, \dots, \vec{r}_N, \vec{p}_1, \dots, \vec{p}_N) = \sum_{i=1}^N \frac{\vec{p}_i^2}{2m_i} + U(\vec{r}_1, \dots, \vec{r}_N)$

$\mathcal{H}(q_1, \dots, q_{3N}, p_1, \dots, p_{3N}) = \frac{1}{2} \sum_\alpha \sum_\beta p_\alpha G_{\alpha\beta}^{-1} p_\beta + U(q_1, \dots, q_{3N})$

Hamilton equations $\dot{q}_\alpha = \frac{\partial \mathcal{H}}{\partial p_\alpha}$ $\dot{p}_\alpha = -\frac{\partial \mathcal{H}}{\partial q_\alpha}$ $\frac{\mathcal{H}}{dt} = 0$ $\mathcal{H} = const$

Some properties

Conservation laws $\frac{da}{dt} = \frac{\partial a}{\partial x_i} \dot{x}(t) = \{a, \mathcal{H}\} = 0$

Incompressibility $\nabla_x(x) = 0$

Symplectic structure $M = J^T M J$ $J_{kl} = \frac{\partial x_k(t)}{\partial x_l(0)}$

Theoretical foundations of statistical mechanics

Thermodynamics

Equilibrium $g(N, P, V, T) = 0$ First law $\Delta E = \Delta Q + \Delta W$

State function $f(n, P, V, T)$ Entropy $\Delta S = \int_1^2 \frac{dQ_{rev}}{T}$

Reversible work $dW_{rev} = -PdV + \mu dN$

Heat $dQ_{rev} = CdT$

The ensemble

Average $A = \frac{1}{Z} \sum_{\lambda=1}^N a(x_\lambda) \equiv \langle a \rangle$

Microstate $x_0 = (q_1(0), \dots, q_{3N}(0), p_1(0), \dots, p_{3N}(0))$

Phase space volume $dx_t = J(x_t; x_0) dx_0$ $\frac{dJ}{dt} = 0 \Rightarrow J(x_t; x_0) = 1 \Rightarrow dx_t = dx_0$

$f(x_t) : \int f(x) dx = 1 \wedge \frac{df(x_t, t)}{dt} = 0 \Rightarrow$

Distribution function $f(x_t, t) dx_t = f(x_0, 0) dx_0 \Rightarrow$

$\frac{\partial f(x, t)}{\partial t} + \{f(x, t), \mathcal{H}(x, t)\} = 0$

Equilibrium $A = \int a(x) f(x, t) dx \Rightarrow \frac{\partial f(x, t)}{\partial t} = 0 \wedge \{f(x, t), \mathcal{H}(x, t) = 0\} \Rightarrow$

$f(x) \propto \mathcal{F}(\mathcal{H}(x))$

$Z = \int dx \mathcal{F}(\mathcal{H}(x)) \Rightarrow f(x) = \frac{1}{Z} \mathcal{F}(\mathcal{H}(x))$

Microcanonical ensemble

State and distribution function

$$\begin{aligned} \text{State function } dS &= \frac{1}{T}dE + \frac{P}{T}dV - \frac{\mu}{T}dN \\ \left(\frac{\partial S}{\partial E}\right)_{V,N} &= \frac{1}{T} \quad \left(\frac{\partial S}{\partial V}\right)_{N,E} = \frac{P}{T} \quad \left(\frac{\partial S}{\partial N}\right)_{V,E} = \frac{\mu}{T} \\ \text{Boltzmann relation } S(N,V,E) &= k \ln \Omega(N,V,E) \\ \Omega(N,V,E) &= M_N \int d\vec{p} \int_{D(V)} d\vec{r} \delta(\mathcal{H}(\vec{r}, \vec{p}) - E) \end{aligned}$$

$$\text{Distribution function} = M_N \int dx \delta(\mathcal{H}(x) - E)$$

$$M_N = \frac{E_0}{N! h^{3N}}$$

where $\frac{1}{h^{3N} N!}$ for undistinguishability

$$\begin{aligned} A = \langle a \rangle &= \frac{M_N}{\Omega(N,V,E)} \int dx a(x) \delta(\mathcal{H}(x) - E) = \frac{\int dx a(x) \delta(\mathcal{H}(x) - E)}{\int dx \delta(\mathcal{H}(x) - E)} \\ \left\langle x_i \frac{\partial \mathcal{H}}{\partial x_j} \right\rangle &= \frac{M_N}{\Omega(N,V,E)} \frac{\partial}{\partial E} \int_{\mathcal{H}(x) < E} dx x_i \frac{\partial \mathcal{H}(x)}{\partial x_j} \end{aligned}$$

Virial theorem

$$\begin{aligned} \left\langle x_i \frac{\partial \mathcal{H}}{\partial x_j} \right\rangle &= \delta_{ij} \frac{\Sigma(E)}{\frac{\partial \Sigma(E)}{\partial E}} \\ \Sigma(N,V,E) &= \frac{1}{N! h^{3N}} \int dx \theta(E - \mathcal{H}) \\ \Omega(N,V,E) &= E_0 \frac{\partial \Sigma(N,V,E)}{\partial E} \quad \left\langle x_i \frac{\partial \mathcal{H}}{\partial x_j} \right\rangle = \delta_{ij} \left(\frac{\ln \Sigma(E)}{\frac{\partial \Sigma(E)}{\partial E}} \right)^{-1} \\ S(N,V,E) &= k \ln \Omega(N,V,E) \simeq k \ln \Sigma(N,V,E) = \tilde{S}(N,V,E) \\ \left\langle x_i \frac{\partial \mathcal{H}}{\partial x_j} \right\rangle &\simeq \delta_{ij} \left(\frac{S(E)}{\frac{\partial S(E)}{\partial E}} \right)^{-1} = kT \delta_{ij} \end{aligned}$$

Thermal contact

$$\begin{aligned} \Omega(N,V,E) &= M_N \int dx \delta(\mathcal{H}_1(x_1) + \mathcal{H}_2(x_2) - E) \\ \Omega(N,V,E) &= \int dE_1 \Omega_1(N_1, V_1, E_1) \Omega_2(N_2, V_2, E - E_1) \\ S(N,V,E) &= k \ln \Omega_1(N_1, V_1, \bar{E}_1) + k \ln \Omega_2(N_2, V_2, E - \bar{E}_1) \\ &= S_1(N_1, V_1, \bar{E}_1) + S_2(N_2, V_2, E - \bar{E}_1) \\ T_1 &= T_2 \end{aligned}$$

Introduction to molecular dynamics

Verlet algorithm

$$\begin{aligned} \vec{r}_i(t + \Delta t) &= 2\vec{r}_i(t) - \vec{r}_i(t - \Delta t) + \frac{\Delta t^2}{m_i} \vec{F}_i(t) \\ \vec{v}_i(t + \Delta t) &= \vec{v}_i(t) + \frac{\Delta t}{2m_i} \left[\vec{F}_i(t) + \vec{F}_i(t + \Delta t) \right] \\ \text{Initial conditions } f(v) &= \sqrt{\frac{m}{2\pi kT}} e^{-\frac{mv^2}{2kT}} \quad f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} \end{aligned}$$

Introduction to molecular dynamics (contd)

Action integral

$$\begin{aligned} Q \equiv \{q_1, \dots, q_{3N}\} \quad \dot{Q} \equiv \{\dot{q}_1, \dots, \dot{q}_{3N}\} \\ A[Q] = \int_{t_1}^{t_2} \mathcal{L}(Q(t), \dot{Q}(t)) dt \\ \delta Q(t_1) = \delta Q(t_2) = 0 \quad \delta \dot{Q}(t_1) = \delta \dot{Q}(t_2) = 0 \\ \delta A = \int_{\alpha=1}^{3N} \frac{\partial \mathcal{L}}{\partial q_\alpha} \delta q_\alpha(t) \Big|_{t_1}^{t_2} dt + \int_{t_1}^{t_2} \sum_{\alpha=1}^{3N} \left[\frac{\partial \mathcal{L}}{\partial q_\alpha} \delta q_\alpha(t) - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_\alpha} \right) \delta q_\alpha(t) \right] dt = 0 \end{aligned}$$

Constraints

$$\begin{aligned} \sum_{\alpha=1}^{3N} a_{k\alpha} dq_\alpha + a_{kt} dt = 0, k = 1, \dots, N_C \\ \text{Holonomic } a_{k\alpha} = \frac{\partial \sigma_k}{\partial q_\alpha} \quad a_{kt} = \frac{\partial \sigma_k}{\partial t} \\ \frac{1}{2} \sum_i m_i \dot{r}_i^2 - C = 0 \Rightarrow \frac{1}{2} \sum_i m_i \dot{r}_i^2 d\vec{r}_i - C dt = 0 \\ \text{Non-holonomic} \Rightarrow a_{1i} = \frac{1}{2} m_i \dot{r}_i \wedge a_{1t} = -C \\ \text{Lagrange multiplier } \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_\alpha} \right) - \frac{\partial \mathcal{L}}{\partial q_\alpha} = \sum_{k=1}^{N_C} \lambda_k a_{k\alpha} \\ \dot{q}_\alpha = \frac{\partial \mathcal{H}}{\partial p_\alpha} \quad \dot{p}_\alpha = -\frac{\partial \mathcal{H}}{\partial q_\alpha} - \sum_{k=1}^{N_C} \lambda_k a_{k\alpha} \quad \sum_{\alpha=1}^{3N} a_{k\alpha} \frac{\partial \mathcal{H}}{\partial p_\alpha} = 0 \\ \text{Simulation } m_i \ddot{\vec{r}}_i = \vec{F}_i + \sum_{k=1}^{N_C} \lambda_k \nabla_i \sigma_k \quad \dot{\sigma}_k = \sum_{i=1}^N \nabla_i \sigma_k \cdot \dot{\vec{r}}_i = 0 \\ \text{Velocity Verlet } \vec{r}_i(\Delta t) = \vec{r}_i(0) + \Delta t \vec{v}_i(0) + \frac{\Delta t^2}{2m_i} \vec{F}_i(0) + \frac{\Delta t^2}{2m_i} \sum_k \lambda_k \nabla_i \sigma_k(0) \\ \vec{r}_i(\Delta t) = \vec{r}_i + \frac{1}{m_i} \sum_k \tilde{\lambda}_k \nabla_i \sigma_k(0) \quad \tilde{\lambda}_k = \frac{\Delta t^2}{2} \lambda_k \\ \sigma_i \left(\vec{r}_1^{(1)}, \dots, \vec{r}_N^{(1)} \right) + \sum_{i=1}^N \sum_{k=1}^{N_C} \frac{1}{m_i} \nabla_i \sigma_k \left(\vec{r}_1^{(1)}, \dots, \vec{r}_N^{(1)} \right) \cdot \nabla_i \sigma_k \left(\vec{r}_1(0), \dots, \vec{r}_N(0) \right) \delta \tilde{\lambda}_k \approx 0 \end{aligned}$$

Direct translation

Liouville operator

$$\begin{aligned} \text{Computable on } a: \frac{da}{dt} = \{a, \mathcal{H}\} \\ iL = \sum_\alpha \left[\frac{\partial \mathcal{H}}{\partial q_\alpha} \frac{\partial}{\partial p_\alpha} - \frac{\partial \mathcal{H}}{\partial p_\alpha} \frac{\partial}{\partial q_\alpha} \right] \Rightarrow iL a = \{a, \mathcal{H}\} \Rightarrow \frac{da}{dt} = iL a \Rightarrow a(x_t) = e^{iL t} a(x_0) \\ \text{Split } iL_1 = \sum_\alpha \frac{\partial \mathcal{H}}{\partial p_\alpha} \frac{\partial}{\partial q_\alpha} \quad iL_2 = -\sum_\alpha \frac{\partial \mathcal{H}}{\partial q_\alpha} \frac{\partial}{\partial p_\alpha} \\ iL_1 iL_2 \phi(x) \neq iL_2 iL_1 \phi(x) \Rightarrow iL_1 iL_2 - iL_2 iL_1 \equiv [iL_1, iL_2] \neq 0 \end{aligned}$$

Trotter theorem

$$\begin{aligned} [iL_1, iL_2] \neq 0 \Rightarrow e^{iL_1 t} e^{iL_2 t} \\ e^{A+B} = \lim_{P \rightarrow \infty} \left[e^{\frac{B}{2P}} e^{\frac{A}{P}} e^{\frac{B}{2P}} \right]^P \quad e^{iL t} = \lim_{P \rightarrow \infty} \left[e^{\frac{iL_1 t}{2P}} e^{\frac{iL_2 t}{P}} e^{\frac{iL_1 t}{2P}} \right]^P \\ e^{iL t} \approx e^{\frac{iL_1 \Delta t}{2}} e^{iL_2 \Delta t} e^{\frac{iL_1 \Delta t}{2}} \end{aligned}$$

Trotter algorithm

$$\begin{aligned} \text{Exponential operator } e^{c \frac{\partial}{\partial x}} g(x) = g(x + c) \\ \begin{pmatrix} x(\Delta t) \\ p(\Delta t) \end{pmatrix} = \begin{pmatrix} x(0) + \frac{\Delta t}{m} \left(p(0) + \frac{\Delta t}{2} F(x(0)) \right) \\ p(0) + \frac{\Delta t}{2} F(x(0)) + \frac{\Delta t}{2} F \left(x(0) + \frac{\Delta t}{m} \left(p(0) + \frac{\Delta t}{2} F(x(0)) \right) \right) \end{pmatrix} \\ x(\Delta t) = x(0) + v(0) \Delta t + \frac{\Delta t^2}{2m} F(0) \quad p(\Delta t) = v(0) + \frac{\Delta t}{2m} [F(0) + F(\Delta t)] \end{aligned}$$

RESPA

$$\begin{aligned} iL = \frac{p}{m} \frac{\partial}{\partial x} + [F_{fast}(x) + F_{slow}(x)] \frac{\partial}{\partial p} = iL_{fast} + iL_{slow} \quad \mathcal{H}_{ref} = \frac{p^2}{2m} + U_{fast}(x) \\ e^{iL \Delta t} = e^{iL_{slow} \frac{\Delta t}{2}} e^{iL_{fast} \Delta t} e^{iL_{slow} \frac{\Delta t}{2}} \\ e^{iL_{fast} \Delta t} = \left[e^{\frac{\Delta t}{2} F_{fast} \frac{\partial}{\partial p}} e^{\delta t \frac{p}{m} \frac{\partial}{\partial x}} e^{\frac{\Delta t}{2} F_{fast} \frac{\partial}{\partial p}} \right]^n \quad \delta t = \frac{\Delta t}{n} \end{aligned}$$

Evaluation of energy and forces

Periodic boundary condition

$$\begin{aligned} \text{Non bonded interaction } U_{nb}(\vec{r}_1, \dots, \vec{r}_N) = \sum_{i < j \in nb} \left\{ 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \frac{q_i q_j}{r_{ij}} \right\} \\ \text{Error function } \text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad \lim_{x \rightarrow \infty} \text{erf}(x) = 1 \quad \text{erf}(0) = 0 \\ \text{Complement error } \text{erfc}(x) = 1 - \text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt \quad \lim_{x \rightarrow \infty} \text{erf}(x) = 1 \\ \frac{1}{r} = \frac{\text{erfc}(\alpha r)}{r} + \frac{\text{erf}(\alpha r)}{r} \quad U_{nb} = U_{short} + U_{long} \quad \vec{r}_{ij} = |\vec{r}_i - \vec{r}_j + \vec{S}| \quad \vec{S} = \vec{m}L \\ U_{short}(\vec{r}_1, \dots, \vec{r}_N) = \sum_{\vec{S}} \sum_{i > j \in nb} \left\{ 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij,S}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij,S}} \right)^6 \right] + \frac{q_i q_j \text{erfc}(\alpha r_{ij,S})}{r_{ij,S}} \right\} \\ U_{long}(\vec{r}_1, \dots, \vec{r}_N) = \sum_{\vec{S}} \sum_{i > h \in nb} \frac{q_i q_h \text{erf}(\alpha r_{ij,S})}{r_{ij,S}} \end{aligned}$$

Short range forces

$$\tilde{U}_{short} = U_{short}(\vec{r}_{ij}) S(\vec{r}_{ij}) \quad S(r) = \begin{cases} 1 & r < r_C - \lambda \\ 1 + \left(\frac{r - r_C + \lambda}{\lambda} \right)^2 (2 \frac{r - r_C + \lambda}{\lambda} - 3) & r_C - \lambda \leq r < r_C \\ 0 & r > r_C \end{cases}$$

Evaluation of energy and forces (contd)

Long range forces

$$\begin{aligned} C_{\vec{g}} = \frac{4\pi}{|\vec{g}|^2} e^{-\frac{|\vec{g}|^2}{4\alpha^2}} \quad \frac{1}{V} \sum_{\vec{g}} C_{\vec{g}} e^{i\vec{g} \cdot \vec{r}} = \frac{1}{V} \sum_{i \neq j} q_i q_j \sum_{\vec{g} \in S} \frac{4\pi}{|\vec{g}|^2} e^{-\frac{|\vec{g}|^2}{4\alpha^2}} e^{i\vec{g} \cdot (\vec{r}_i - \vec{r}_j)} \\ U_{long} = \frac{1}{V} \sum_{i,j} q_i q_j \sum_{\vec{g} \in S} \frac{4\pi}{|\vec{g}|^2} e^{-\frac{|\vec{g}|^2}{4\alpha^2}} e^{i\vec{g} \cdot (\vec{r}_i - \vec{r}_j)} - \frac{1}{V} \sum_i q_i^2 \sum_{\vec{g} \in S} \frac{4\pi}{|\vec{g}|^2} e^{-\frac{|\vec{g}|^2}{4\alpha^2}} \\ U_{long} = \frac{1}{V} \sum_{\vec{g} \in S} \frac{4\pi}{|\vec{g}|^2} e^{-\frac{|\vec{g}|^2}{4\alpha^2}} |S(\vec{g})|^2 - \frac{\alpha}{\sqrt{\pi}} \sum_i q_i^2 \end{aligned}$$

Particle-particle particle-mesh Ewald

$$\begin{aligned} \rho(\vec{r}) = \sum_i q_i \delta(\vec{r} - \vec{r}_i) \quad \rho(\vec{g}) = \int d\vec{r} \rho(\vec{r}) e^{i\vec{g} \cdot \vec{r}} = \sum_i q_i e^{i\vec{g} \cdot \vec{r}_i} \\ \nabla^2 \phi(\vec{r}) = -\nabla \cdot \vec{E} = -4\pi \rho(\vec{r}) \quad g^2 \phi(\vec{g}) = -4\pi \rho(\vec{g}) = 4\pi S(\vec{g}) \end{aligned}$$

Canonical ensemble

Thermodynamics

$$\begin{aligned} \text{Helmholtz free energy } A(N,V,T) = E(N,V,T) - TS(N,V,T) \\ dA = SdT - PdV + \mu dN \quad S = -\left(\frac{\partial A}{\partial T}\right)_{N,V} \quad P = -\left(\frac{\partial A}{\partial V}\right)_{N,T} \quad \mu = \left(\frac{\partial A}{\partial N}\right)_{V,T} \end{aligned}$$

Thermal contact

$$\begin{aligned} \text{Microcanonical } \Omega(N,V,E) = M_N \int dx_1 dx_2 \delta(\mathcal{H}_1(x_1) + \mathcal{H}_2(x_2) - E) \\ \text{Distribution function } \ln f(x_1) = \ln \int dx_2 \delta(\mathcal{H}_1(x_1) + \mathcal{H}_2(x_2) - E) \\ \ln f(x_1) \approx \ln \int dx_2 \delta(\mathcal{H}_2(x_2) - E) - \frac{\partial}{\partial E} \ln \int dx_2 \delta(\mathcal{H}_2(x_2) - E) \mathcal{H}_1(x_1) \\ Q(N,V,T) = \frac{1}{N! h^{3N}} \int dx e^{-\beta \mathcal{H}(x)} = \sum_i g_i e^{\beta E_i} \quad \beta = \frac{1}{kT} \\ P(E) dE = \int_0^\infty e^{-\beta E} \frac{dE g(E)}{g(E) dE} \end{aligned}$$

Connection canonical microcanonical

While in the case of a microcanonical ensemble the energy in the system is fixed, in the canonical case it is not. Imagine that you have multiple molecules; if the total energy is not fixed, then the partition function is calculated taking into account the Boltzmann weights of the possible energy levels. On the contrary, in the microcanonical case, the energy is fixed and only a certain combination of configurations will be allowed.

From micro to macro

$$\begin{aligned} A = E - \beta \left(\frac{\partial A}{\partial \beta} \right)_{N,V} \\ E = \langle \mathcal{H} \rangle = \frac{1}{N! h^{3N}} \frac{\int dx \mathcal{H}(x) e^{-\beta \mathcal{H}(x)}}{\int dx e^{-\beta \mathcal{H}(x)}} = -\frac{1}{Q(N,V,\beta)} \frac{\partial Q(N,V,\beta)}{\partial \beta} = -\frac{\partial \ln Q(N,V,\beta)}{\partial \beta} \\ A + \frac{\partial \ln Q}{\partial \beta} + \beta \frac{\partial A}{\partial \beta} = 0 \Rightarrow \ln Q(N,V,\beta) = -\beta A(N,V,\beta) \\ A(N,V,T) = -kT \ln Q(N,V,T) \quad C_N = \frac{1}{N! h^{3N}} \\ \text{Energy } E = \langle \mathcal{H} \rangle = \frac{1}{Q} \frac{\partial Q}{\partial \beta} = 0 \\ \text{Temperature estimator } \mathcal{T}(x) + \frac{1}{3Nk} \sum_i \frac{p_i^2}{m_i} \quad T = \langle \mathcal{T}(x) \rangle = \frac{C_N \int dx \mathcal{T}(x) e^{-\beta \mathcal{H}(x)}}{C_N \int dx e^{-\beta \mathcal{H}(x)}} \\ \text{Energy fluctuation } \Delta E^2 = \frac{\partial^2 \ln Q}{\partial \beta^2} = kT^2 C_V \quad \frac{\Delta E}{E} \sim \frac{1}{\sqrt{N}} \\ \text{Pressure estimator } \mathcal{P}(\vec{r}, \vec{p}) = \frac{1}{3V} \sum_i \left[\frac{p_i^2}{m_i} + \vec{F}_i \cdot \vec{r}_i \right] \end{aligned}$$

Other information

When free particle moving on a direction $Q = \left(\frac{2\pi m}{h^2} \right)^{1/2} X$ with thermal wavelength $\Lambda = h \left(\frac{\beta}{2\pi m} \right)^{1/2}$

Special cases

When series of states separated by infinite number of equally spaced nondegenerate energy levels $Q = \frac{1}{1 - e^{-\beta \epsilon}}$. In this situation $p_i = \frac{e^{-\beta \epsilon_i}}{q} = (1 - e^{-\beta \epsilon}) e^{-\beta \epsilon_i}$
collection of N independent molecules
 Energy: $E_i = \epsilon_i(1) + \epsilon_i(2) + \dots + \epsilon_i(N)$
 Partition function: $Q = \sum_i e^{-\beta \epsilon_i(1) - \beta \epsilon_i(2) - \dots - \beta \epsilon_i(N)} = \left(\sum_i e^{-\beta \epsilon_i} \right)^N = q^N$
 $\frac{q^N}{N!}$ if undistinguishable particles

Thermostats

Velocity rescaling

$$\begin{aligned} \bar{K} = \frac{N_j}{2\beta} \quad K = \frac{1}{2} \sum_i m_i \bar{v}_i^2 \quad \bar{v}_i \rightarrow \frac{\bar{v}_i}{\alpha} \quad \alpha = \sqrt{\frac{\bar{K}}{K}} \\ \bar{K} = \frac{1}{2} \sum_i m_i \frac{\bar{v}_i^2}{\alpha^2} = \frac{\bar{K}}{2K} \sum_i m_i \bar{v}_i^2 \end{aligned}$$

Andersen-Heyes Thermostat

$$\begin{aligned} P(p) = \left(\frac{\beta}{2\pi m} \right)^{\frac{3}{2}} e^{-\beta \frac{p^2}{2m}} \\ P(K_t) dK_t \propto K_t^{\frac{N_f}{2} - 1} e^{-\beta K_t} dK_t \\ K_t = \frac{1}{2} \sum_i m_i \frac{\bar{v}_i^2}{\alpha^2} = \frac{K_t}{2K} \sum_i m_i \bar{v}_i^2 \end{aligned}$$

Langevin Thermostat

$$\begin{aligned} \dot{\vec{r}}_i(t) = \beta D \vec{F}_i(\vec{r}_1(t), \dots, \vec{r}_N(t)) + \eta_i(t) \quad \frac{\partial}{\partial t} P(\vec{x}, t) = D \sum_{i=1}^N \nabla_i \cdot [\nabla_i + \beta \nabla \mathcal{H}(\vec{x})] \\ P(\vec{x}) = C e^{-\beta \mathcal{H}(\vec{x})} \quad \langle \eta_i(t) \eta_j(t) \rangle = 6D \delta_{ij} \frac{\delta \lambda}{\Delta t} \quad \sigma^2 = \frac{6D}{\Delta t} \\ \xi_i(t_k) = \sqrt{\frac{\Delta t}{6\beta}} \eta_i(t_k) \quad \langle \xi_i(t_k) \rangle = 0 \quad \langle \xi_i(t_k) \xi_j(t_l) \rangle = \delta_{ij} \delta_{kl} \\ \vec{r}_i(t + \Delta t) = \vec{r}_i(t) + \Delta t \frac{\vec{F}_i(\vec{r}_1(t), \dots, \vec{r}_N(t))}{m_i \gamma} + \sqrt{\frac{6kT \Delta t}{m_i \gamma}} \xi_i(t) \\ j \end{aligned}$$

Bussi velocity Verlet

$$\begin{aligned} dK = \left(D(K) \frac{\partial \log P(K)}{\partial K} + \frac{\partial D(K)}{\partial K} \right) dt + \sqrt{2D(K)} dW \quad P(K_t) dK_t \propto K_t^{\frac{N_f}{2} - 1} e^{-\beta K_t} dK_t \\ dK = \left(\frac{N_f D(K)}{2KK} (K - \bar{K}) - \frac{D(K)}{K} + \frac{\partial D(K)}{\partial K} \right) dt + \sqrt{2D(K)} dW \quad D(K) = \frac{2KK}{N_f \tau} \\ dK = (K - \bar{K}) \frac{dt}{\tau} + \sqrt{\frac{2KK}{N_f}} \frac{dW}{\sqrt{\tau}} \\ \text{Berendsen's thermostat } dK = (K - \bar{K}) \frac{dt}{\tau} \\ \alpha^2 = e^{-\frac{\Delta t}{\tau}} + \frac{\bar{K}}{N_f K} \left(1 - e^{-\frac{\Delta t}{\tau}} \right) \sum_{i=1}^{N_f} R_i^2 + 2e^{-\frac{\Delta t}{\tau}} \sqrt{\frac{\bar{K}}{N_f K} \left(1 - e^{-\frac{\Delta t}{\tau}} R_1 \right)} \end{aligned}$$

Nosé Hamiltonian

$$\begin{aligned} \mathcal{H}_N = \sum_i \frac{\vec{p}_i^2}{2m_i s^2} + U(\vec{r}_1, \dots, \vec{r}_n) + \frac{p_s^2}{2Q} + gkT \log s \\ \Omega = \int d^N \vec{r} d^N \vec{p} ds dp_s dN \delta \left(\mathcal{H} + \frac{p_s^2}{2Q} + gkT \log s - E \right) \\ \Omega = \frac{1}{dkT} \int d^N \vec{r} d^N \vec{p} dp_s e^{\frac{dN+1}{gkT} (E - \mathcal{H} - \frac{p_s^2}{2Q})} \\ g = dN + 1 \Rightarrow \Omega = \frac{e^{\frac{1}{kT} \sqrt{2\pi QkT}}}{(dN+1)kT} \int d^N \vec{r} d^N \vec{p} e^{-\frac{\mathcal{H}}{kT}} \\ \dot{\vec{r}}_i = \frac{\vec{p}_i}{m_i} s^2 \quad \dot{\vec{p}}_i = \vec{F}_i \quad \dot{s} = \frac{p_s}{Q} \quad \dot{p}_s = \frac{1}{s} \left[\sum_i \frac{\vec{p}_i^2}{m_i s^2} - gkT \right] \end{aligned}$$

Thermostats (contd)

Nosé-Hoover equations

$$\begin{aligned} \dot{\vec{r}}_i = \frac{\vec{p}_i}{m_i} \quad \dot{\vec{p}}_i = \frac{\vec{p}_i}{s} \quad \frac{d\vec{r}_i}{dt} = \frac{\vec{p}_i}{m_i} \quad \frac{d\vec{p}_i}{dt} = \vec{F}_i - \frac{s p_i^2}{Q} \vec{r}_i \\ \frac{ds}{dt} = \frac{s^2 p_s}{Q} \quad \frac{dp_s}{dt} = \frac{1}{2} \left[\sum_i \left(\frac{\vec{p}_i^2}{m_i} \right) - gkT \right] - \frac{s (p_s')^2}{Q} \quad \frac{1}{2} \frac{ds}{dt} = \frac{dn}{dt} \quad p_s = p_\eta = s p_s' \\ \dot{\vec{r}}_i = \frac{\vec{p}_i}{m_i} \quad \dot{\vec{p}}_i = \vec{F}_i - \frac{p_\eta}{Q} \vec{p}_i \quad \dot{\eta} = \frac{p_\eta}{Q} \quad \dot{p}_\eta = \sum_i \frac{\vec{p}_i^2}{m_i} - dNkT \end{aligned}$$

Non Hamiltonian statistical mechanics

$$\begin{aligned} \dot{x} = \xi(x, t) \quad \nabla \cdot \dot{x} = \nabla \cdot \xi(x, t) = \kappa(x, t) \neq 0 \\ J(x_t; x_0) = e^{\int_{x_0}^{x_t} ds \kappa(s, s)} \quad \kappa(x_t, t) = \frac{dw(x_t, t)}{dt} \Rightarrow J(x_t; x_0) = e^{\int_{x_0}^{x_t} dw(x_t, t) - w(x_0, 0)} = e^{-w(x_t, 0)} dx_0 \\ J(x_t; x_0) = \frac{\sqrt{g(x_0, 0)}}{\sqrt{g(x_t, t)}} \quad \sqrt{g(x_t, t)} = e^{-w(x_t, t)} \\ \frac{\partial}{\partial t} [f(x, t) \sqrt{g(x, t)}] + \nabla \cdot [x \sqrt{g(x, t)} f(x, t)] = 0 \quad f(x, t) \sqrt{g(x, t)} dx_t = f(x_0) \sqrt{g(x_0)} dx_0 \quad \xi(x) \cdot \nabla f(x) = 0 \\ \Lambda_k(x_t) - C_k = 0 \quad \frac{d\Lambda_k(x_t)}{dt} = 0 \Rightarrow f(x) = \prod_{k=1}^{N_C} \delta(\Lambda_k(x_t) - C_k) \end{aligned}$$

Microcanonical $\mathcal{E} = \int dx \sqrt{g(x)} f(x) = \int dx \sqrt{g(x)} \prod_{k=1}^{N_C} \delta(\Lambda_k(x_t) - C_k)$

$$\begin{aligned} \text{Nosé-Hoover: } \mathcal{H}'(\vec{r}, \eta, \vec{p}, p_\eta) = \mathcal{H}(\vec{r}, \vec{p}) + \frac{p_\eta^2}{2Q} + dNkT\eta \quad \frac{d\mathcal{H}'}{dt} = 0 \\ \kappa = -N d\eta \Rightarrow \sqrt{g} = e^{dN\eta} \\ \text{Partition function } \mathcal{E}_T(N, V, C_1) = \frac{e^{\beta C_1} \sqrt{2\pi QkT}}{dNkT} \int d^N \vec{p} \int_{D(V)} d^N \vec{r} e^{-\beta \mathcal{H}'(\vec{r}, \vec{p})} \\ \vec{P} = \sum_{i=1}^N \vec{p}_i \quad \vec{K} = \vec{P} e^\eta \Rightarrow \frac{d\vec{K}}{dt} = 0 \end{aligned}$$

Nosé-Hoover chains

$$\begin{aligned} \dot{\vec{r}}_i = \frac{\vec{p}_i}{m_i} \quad \dot{\vec{p}}_i = \vec{F}_i - \frac{p_{\eta_1}}{Q_1} \vec{p}_i \quad \dot{\eta}_j = \frac{p_{\eta_j}}{Q_j} \quad j = 1, \dots, M \\ \dot{p}_{\eta_1} = \left[\sum_i \frac{\vec{p}_i^2}{m_i} - dNkT \right] - \frac{p_{\eta_2}}{Q_2} p_{\eta_1} \quad \dot{p}_{\eta_j} = \left[\sum_i \frac{\vec{p}_{\eta_{j-1}}^2}{Q_{j-1}} - kT \right] - \frac{p_{\eta_{j+1}}}{Q_{j+1}} p_{\eta_j} \quad j = 2, \dots, M-1 \\ \dot{p}_{\eta_M} = \frac{p_{\eta_{M-1}}^2}{Q_{M-1}} - kT \\ \mathcal{H}'(\vec{r}, \eta, \vec{p}, p_\eta) = \mathcal{H}(\vec{r}, \vec{p}) + \sum_{j=1}^M \frac{p_{\eta_j}^2}{2Q_j} + dNkT\eta_1 + kT \sum_{j=2}^M \eta_j \\ \kappa = -dN \dot{\eta}_1 - \dot{\eta}_c \quad \eta_c = \sum_{j=2}^M \eta_j \quad \sqrt{g} = e^{dN \eta_1 + \eta_c} \\ \text{Partition function } \mathcal{E}_T(N, V, C_1) = \mathcal{M} \int d^N \vec{p} \int_{D(V)} d^N \vec{r} e^{-\beta \mathcal{H}'(\vec{r}, \vec{p})} \end{aligned}$$

Isobaric ensemble

Legendre transforms

$$\begin{aligned} \text{Enthalpy } dH = TdS + \mu dN + VdP \quad T = \left(\frac{\partial H}{\partial S} \right)_{N,P} \quad \langle V \rangle = \left(\frac{\partial H}{\partial P} \right)_{N,S} \quad \mu = \left(\frac{\partial H}{\partial N} \right)_{P,S} \\ \text{Gibbs } dG = \mu dN + VdP - SdT \quad S = -\left(\frac{\partial G}{\partial T} \right)_{N,P} \quad \langle V \rangle = -\left(\frac{\partial G}{\partial P} \right)_{N,T} \quad \mu = -\left(\frac{\partial G}{\partial N} \right)_{P,T} \end{aligned}$$

Isoenthalpic-isobaric ensemble

$$\begin{aligned} H = \mathcal{H}(v) + PV \quad f(x) = F(\mathcal{H}(x)) = \mathcal{M} \delta(\mathcal{H}(x) + PV - H) \\ \Gamma(N, P, H) = \mathcal{M} \int_0^\infty dV \int d^N \vec{p} \int_{D(V)} d^N \vec{r} \delta(\mathcal{H}(\vec{r}, \vec{p}) + PV - H)$$

Euler's theorem

$$U(\lambda N, \lambda V, \lambda S) = \lambda U(N, V, S) \Rightarrow U(N, V, S) = \mu N - PV + TS$$

$$A(\lambda N, \lambda V, T) = \lambda A(N, V, T) \Rightarrow A(N, V, T) = \mu N - PV$$

$$H(\lambda N, \lambda S, P) = \lambda H(N, S, P) \Rightarrow H(N, S, P) = \mu N + TS$$

$$G(\lambda N, P, T) = \lambda G(N, P, T) \Rightarrow G(N, P, T) = \mu N$$

Thermodynamics

$$\tau(A) \left(\frac{\partial A}{\partial N}, V, T \right) = -PV$$

$$\dot{A}(\mu, V, T) = A(N(\mu), V, T) - \mu N(\mu) \quad d\dot{A} = -PdV - SdT - Nd\mu$$

$$\mathcal{E}(\mu, V, T) = \sum_{N=0}^{\infty} \frac{1}{N! h^{3N}} e^{\beta \mu N} \int dx e^{-\beta \mathcal{H}(x, N)} = \sum_{N=0}^{\infty} e^{\beta \mu N} Q(N, V, T) = e^{\beta PV}$$

$$\frac{PV}{kT} = \ln \mathcal{E}(\mu, V, T) \quad z(\text{fugacity}) = e^{\beta \mu} \quad \mathcal{E}(z, V, T) = \sum_{N=0}^{\infty} z^N Q(N, V, T)$$

Grand potential: $\tilde{A} = kT \ln \mathcal{E}(\mu, T, V)$

$$\langle N \rangle = \frac{1}{\mathcal{E}(\mu, V, T)} \sum_{N=0}^{\infty} N e^{\beta \mu N} Q(N, V, T) = kT \left(\frac{\partial}{\partial \mu} \ln \mathcal{E}(\mu, V, T) \right)_{V, T}$$

$$\langle N \rangle = z \frac{\partial}{\partial z} \ln \mathcal{E}(z, V, T)$$

Ideal gas

$$Q(N, V, T) = \frac{1}{N!} \left[V \left(\frac{2\pi m}{\beta h^2} \right)^{\frac{3}{2}} \right]^N = \frac{1}{N!} \left(\frac{V}{\lambda^3} \right)^N \quad \mathcal{E}(z, V, T) = e^{zV} \lambda^3$$

$$\langle N \rangle = V \frac{z}{\lambda^3} \Rightarrow z = \frac{\langle N \rangle \lambda^3}{V} \quad \frac{PV}{kT} = \ln \mathcal{E}(z, V, T) = \frac{Vz}{\lambda^3} = \langle N \rangle$$

Particle number fluctuations

$$\Delta N = \sqrt{\langle N^2 \rangle - \langle N \rangle^2} \quad z \frac{\partial}{\partial z} z \frac{\partial}{\partial z} \ln \mathcal{E}(z, V, T) = \langle N^2 \rangle - \langle N \rangle^2$$

$$\Delta N^2 = kTV \frac{\partial^2 P}{\partial \mu^2} \quad a(v, T) = \frac{1}{N} A \left(N, \frac{V}{N}, T \right) \Rightarrow \mu = a(v, T) - V \frac{\partial a}{\partial v}$$

$$P = -\frac{\partial a}{\partial v} \quad \frac{\partial P}{\partial \mu} = -\frac{\partial^2 a}{\partial v^2} \frac{\partial v}{\partial \mu} = \frac{\partial \mu}{\partial v} - \frac{\partial a}{\partial v} - v \frac{\partial^2 a}{\partial v^2} \Rightarrow \frac{\partial P}{\partial \mu} = \frac{1}{v}$$

$$\frac{\partial^2 P}{\partial \mu^2} = -\frac{1}{v^3} \frac{\partial P}{\partial v} \quad \kappa_T = -\frac{1}{v} \frac{\partial P}{\partial \mu} \Rightarrow \Delta N^2 = \frac{\langle N \rangle kT}{v} \kappa_T \quad \frac{\Delta N}{\langle N \rangle} \sim \frac{1}{\sqrt{\langle N \rangle}}$$

Quantifying uncertainties and sampling quality

Autocorrelation analysis

$$C_f(t') = \frac{\langle (f(x) - \langle f \rangle)(f(t+t') - \langle f \rangle) \rangle}{\sigma_f^2}$$

$$C_f(t') = \frac{1}{\sigma_f^2} \frac{1}{N} \sum_{i=1}^{N-t'} (f(j\Delta t) - \langle f \rangle)(f(j\Delta t + t') - \langle f \rangle) \quad \tau_f = \int_0^{+\infty} dt' C_f(t')$$

$$N_f^{ind} \simeq \frac{t_{sim}}{\tau_f} \quad SE(f) = \frac{\sigma_f}{\sqrt{N_f^{ind}}} \simeq \sigma_f \sqrt{\frac{\tau_f}{t_{sim}}}$$

Block averaging analysis

$$BSE(f, n) = \frac{\sigma_n}{\sqrt{M}}$$

Protein motions

Gaussian network models

$$U_{ij} = \gamma_{ij} (\Delta \vec{r}_j - \Delta \vec{r}_i) \cdot (\Delta \vec{r}_j - \Delta \vec{r}_i) = \gamma_{ij} \Delta r_{ij}^2 \quad U_{GNM} = \frac{\gamma}{2} \sum_i \sum_j \Delta r_{ij}^2$$

$$U_{GNM} = \frac{\gamma}{2} \Delta \vec{r}(t)^T \Gamma \Delta \vec{r}(t) \quad \langle \Delta \vec{r}_i \cdot \Delta \vec{r}_j \rangle = \frac{3kT}{\gamma} [\Gamma^{-1}]_{ij}$$

$$\langle \Delta \vec{r}_i \cdot \Delta \vec{r}_i \rangle = \sum_k [\Delta r_i^2]_k \quad B_i = \frac{8\pi}{3} \langle \Delta r_i^2 \rangle$$

Normal mode analysis: $U = \frac{1}{2} \sum_i \sum_j H_{ij} (q_i - q_i^0)(q_j - q_j^0)$

$$C = \langle \Delta \vec{q} \Delta \vec{q}^T \rangle = \frac{1}{Q} \int \Delta \vec{q} \Delta \vec{q}^T e^{-\frac{\Delta \vec{q}^T H \Delta \vec{q}}{2kT}} d^N \Delta \vec{q} = kT H^{-1}$$

Anisotropic network model

$$U_{ANM} = \frac{1}{2} \sum_{ij} \gamma (r_{ij} - r_{ij}^0)^2 \Rightarrow \frac{\partial^2 U}{\partial x_i \partial x_j} = -\gamma \frac{(x_i - x_j)(y_i - y_j)}{r_{ij}^3} \quad \langle \Delta \vec{r}_i \cdot \Delta \vec{r}_j \rangle = \frac{3kT}{\gamma} \sum_k \lambda_k^{-1} [\vec{u}_k \vec{u}_k^T]_{ij}$$

Correlation cosine $I_k = \frac{\Delta \vec{q}_{AB} \cdot \vec{u}_k}{|\Delta \vec{q}_{AB}|} \quad C_0 = \sqrt{\sum_k I_k^2}$

Degree of connectivity $\kappa_k = N^{-1} e^{-\sum_{i=1}^N \alpha (\Delta r_i)^2} \log(\alpha (\Delta r_i)^2|_k) \quad \sum_{i=1}^N \alpha (\Delta r_i)^2|_k = 1$

Essential dynamics

Covariance matrix $C_{ij} = \overline{(x_i(t) - \overline{x_i(t)})(x_j(t) - \overline{x_j(t)})}$

Correlation matrix $R_{ij} = \frac{(x_i(t) - \overline{x_i(t)})(x_j(t) - \overline{x_j(t)})}{\sigma_{x_i} \sigma_{x_j}}$

$$PC_k(t) = \sum_{i=1}^N \vec{d}_i(t) \cdot \vec{u}_i^k \quad c_k = \frac{2}{T_{sim}} \left(\int_0^{T_{sim}} \cos \left(\pi \frac{k_B T}{\lambda_k} t \right) PC_k(t) dt \right)^2 \left(\int_0^{T_{sim}} PC_k^2(t) dt \right)^{-1}$$

$$\Omega_{A,B} = 1 - \left[\frac{\sum_{k=1}^{3N-6} (\lambda_k^A + \lambda_k^B) - 2 \sum_{k=1}^{3N-6} \sum_{j=1}^{3N-6} \sqrt{\lambda_k^A \lambda_j^B} (\vec{u}_k^A \cdot \vec{u}_j^B)^2}{\sum_{k=1}^{3N-6} (\lambda_k^A + \lambda_k^B)} \right]^{\frac{1}{2}}$$

Monte Carlo methods

Introduction

$$f(x) \geq 0 \wedge \int f(x) dx = 1 \quad I = \int dx \phi(x) f(x) \equiv \langle \phi \rangle_f$$

$$\tilde{I}_M = \frac{1}{M} \int_{i=1}^M \phi(x_i) \quad \lim_{M \rightarrow \infty} \tilde{I}_M = I$$

$$\int dx \phi(x) f(x) = \frac{1}{M} \int_{i=1}^M \phi(x_i) \pm \frac{1}{\sqrt{M}} \left[(\phi^2)_f - \langle \phi \rangle_f^2 \right]^{\frac{1}{2}}$$

$$P(X) = \int_a^X f(x) dx \quad f(X) = \frac{dP}{dX} \quad X \geq x \Rightarrow g(X) \geq g(x) \quad \tilde{P}(Y = g(X)) = P(X)$$

Monte Carlo methods (contd)

Importance sampling

$$I = \int dx \phi(x) f(x) = \int dx \left[\frac{\phi(x) f(x)}{h(x)} \right] h(x) = \int dx \psi(x) h(x) = \frac{1}{M} \sum_{i=1}^M \psi(x_i) \pm \frac{1}{\sqrt{M}} \left[\langle \psi^2 \rangle_h - \langle \psi \rangle_h^2 \right]^{\frac{1}{2}}$$

$$\sigma^2[h] = \int dx \frac{\phi^2(x) f^2(x)}{h(x)} - \left[\int dx \phi(x) f(x) \right]^2$$

$$\delta F[h] = -\frac{\phi^2(x) f^2(x)}{h^2(x)} \delta h(x) - \lambda \delta h(x) \quad \frac{\delta F[h]}{\delta h(x)} = 0 \Rightarrow h(x) = \frac{1}{\sqrt{-\lambda}} \phi(x) f(x)$$

Markov chains

Detailed balance $R(x|y)f(y) = R(y|x)f(x)$

Rejection method $R(x|y) = A(x|y)T(x|y) \quad r(x|y) = \frac{T(y|x)f(x)}{T(x|y)f(y)}$

$$A(x|y) = \min[1, r(x|y)]$$

Metropolis algorithm $r(x_{k+1}|x_k) = \frac{T(x_k|x_{k+1})f(x_{k+1})}{T(x_{k+1}|x_k)f(x_k)}$

Canonical distribution $A(r'|r) = \min \left[1, e^{-\beta(U(r') - U(r))} \right]$

$$\begin{cases} x'_i = x_i + \frac{1}{\sqrt{3}}(\xi_x - 0.5)\Delta \\ y'_i = y_i + \frac{1}{\sqrt{3}}(\xi_y - 0.5)\Delta \\ z'_i = z_i + \frac{1}{\sqrt{3}}(\xi_z - 0.5)\Delta \end{cases}$$

Isothermal-isobaric $A(V'|V) = \min \left[1, e^{-\beta P(V' - V)} e^{N \ln \frac{V'}{V}} e^{-\beta(U(r') - U(r))} \right]$

Gran canonical $\begin{cases} A(N+1|N) = \min \left[1, \frac{V}{\lambda^3(N+1)} e^{\beta \mu} e^{-\beta(U(r') - U(r))} \right] \\ A(N-1|N) = \min \left[1, \frac{\lambda^3 N}{V} e^{-\beta \mu} e^{-\beta(U(r') - U(r))} \right] \end{cases}$

Hybrid Monte Carlo

$$A(r', p'|r, p) = \min \left[1, e^{-\beta \Delta \mathcal{H}} \right] \quad T(r', p'|r, p) = T(r, -p|r', -p')$$

$$\int d^N p d^N p' T(r', p'|r, p) A(r', p'|r, p) f(r, p) = \int d^N p d^N p' T(r, p|r', p') A(r, p|r', p') f(r', p')$$

Free energy calculations

Free energy perturbation theory

$$\Delta A_{AB} = -kT \ln \frac{Z_B}{Z_A} \quad Z_A = \langle e^{-\beta[U_B(\vec{r}_1, \dots, \vec{r}_N) - U_A(\vec{r}_1, \dots, \vec{r}_N)]} \rangle_A$$

$$\Delta A_{AB} = -kT \ln \langle e^{-\beta[U_B(\vec{r}_1, \dots, \vec{r}_N) - U_A(\vec{r}_1, \dots, \vec{r}_N)]} \rangle_A$$

Adiabatic switching: $\frac{kT}{Z} \frac{\partial Z}{\partial \lambda} = \frac{kT}{Z} \int d^N \vec{r} \left(-\beta \frac{\partial U}{\partial \lambda} \right) e^{-\beta U(\vec{r}_1, \dots, \vec{r}_N)} = -\langle \frac{\partial U}{\partial \lambda} \rangle$

Thermodynamics integration: $\Delta A_{AB} = \int_0^1 \langle \frac{\partial A}{\partial \lambda} \rangle_{\lambda} d\lambda = \int_0^1 \langle \frac{\partial U}{\partial \lambda} \rangle_{\lambda} d\lambda$

Adiabatic free energy dynamics: $\langle \frac{p_{\lambda}^2}{2m_{\lambda}} \rangle = kT_{\lambda} \quad Z(\lambda, \beta) = \int d^N \vec{r} e^{-\beta U(\vec{r}, \lambda)}$

$$\mathcal{H}_{eff}(\lambda, p_{\lambda}) = \frac{p_{\lambda}^2}{2m_{\lambda}} - \frac{1}{\beta} \ln Z(\lambda, \beta) \quad \tilde{P}_{adb}(\lambda, \beta, p_{\lambda}) = [Z(\lambda, \beta)]^{\frac{\beta}{2}}$$

$$A(\lambda) = -kT \ln Z(\lambda, \beta) + const$$

Jarzynski's equality

$$W_{AB} = \langle \mathcal{W}_{AB}(x) \rangle_A \geq \Delta A_{AB} \quad \langle e^{-\beta \mathcal{W}_{AB}(x_0)} \rangle_A = e^{-\beta \Delta A_{AB}}$$

Pulling experiment: $U(\vec{r}_1, \dots, \vec{r}_N, t) = U_0(\vec{r}_1, \dots, \vec{r}_N) + \frac{1}{2}k(|\vec{r}_1 - \vec{r}_N| - r_{eq} - vt)^2$

$$\langle e^{-\beta \mathcal{W}_{\tau}} \rangle = \int \mathcal{W}_{\tau} P(\mathcal{W}_{\tau}) e^{-\beta \mathcal{W}_{\tau}}$$

$$P(\mathcal{W}_{\tau}) \sim N \Rightarrow \ln \langle e^{-\beta \mathcal{W}_{\tau}} \rangle \simeq -\beta \langle \mathcal{W}_{\tau} \rangle + \frac{\beta^2}{2} (\langle \mathcal{W}_{\tau}^2 \rangle - \langle \mathcal{W}_{\tau} \rangle^2)$$

Replica exchange Monte Carlo

$$F(\vec{r}^{(1)}, \dots, \vec{r}^{(M)}) = \prod_{K=1}^M f_K(\vec{r}^{(K)}) \quad f_K(\vec{r}^{(K)}) = \frac{e^{-\beta_K U(\vec{r}^{(K)})}}{Q(N, V, T_K)}$$

$$T(\vec{r}^{(K)}, \vec{r}^{(K+1)} | \vec{r}^{(K)}, \vec{r}^{(K+1)}) = T(\vec{r}^{(K)}, \vec{r}^{(K+1)} | \vec{r}^{(K+1)}, \vec{r}^{(K)})$$

$$A(\vec{r}^{(K+1)}, \vec{r}^{(K)} | \vec{r}^{(K)}, \vec{r}^{(K+1)}) = \min \left[1, e^{-\Delta_{K, K+1}} \right]$$

$$\Delta_{K, K+1} = (\beta_K - \beta_{K+1}) [U(\vec{r}^{(K+1)}) - U(\vec{r}^{(K)})]$$

Wang-Landau sampling $Q(\beta) = \int_0^{\infty} dE e^{-\beta E} \Omega(E) \quad A(E_2|E_1) = \min \left[1, \frac{\Omega(E_2)}{\Omega(E_1)} \right]$

Rare events

Reaction coordinates

$$P(s_1, \dots, s_n) = \frac{C_N}{Q(N, V, T)} \int d^N \vec{p} \int_{q(\vec{r})=q^{\dagger}} d^N \vec{r} e^{-\beta \mathcal{H}(\vec{r}, \vec{p})} \prod_{\alpha=1}^n \delta(f_{\alpha}(\vec{r}_1, \dots, \vec{r}_N) - s_{\alpha})$$

$$A(s_1, \dots, s_n) = -fT \ln P(s_1, \dots, s_n)$$

Blue moon ensemble

$$\frac{dA}{ds} = -\frac{kT}{P(s)} \frac{dP}{ds} \quad A(q) = A(s^{(i)}) + \int_{s^{(i)}}^q \frac{dA}{ds} ds \quad \Delta A = \int_{s^{(i)}}^{s^{(f)}} \frac{dA}{ds} ds$$

$$P(s) = \langle \delta(f_1(\vec{r}_1, \dots, \vec{r}_N) - s) \rangle \quad \frac{1}{P(s)} \frac{dP}{ds} = -\beta \left\langle \frac{\partial \tilde{H}}{\partial q_1} \right\rangle^{cond}$$

$$A(q) = A(s^{(i)}) + \int_{s^{(i)}}^q \left\langle \frac{\partial \tilde{H}}{\partial q_1} \right\rangle_s^{cond} ds$$

WHAM

Umbrella sampling $W_k(f_1(\vec{r}_1, \dots, \vec{r}_N), s_k) = \frac{1}{k} [f_1(\vec{r}_1, \dots, \vec{r}_N) - s_k]^2 \Rightarrow P(s, s^{(k)})$

$$\tilde{P}(q, s^{(k)}) = e^{\beta A_k} \int d^N \vec{r} e^{-\beta U(\vec{r})} e^{-\beta W_k(f_1(\vec{r}), s^{(k)})} \delta(f_1(\vec{r}) - q)$$

$$e^{-\beta A_k} = e^{-\beta A_0} \left\langle e^{-\beta W_k(f_1(\vec{r}), s^{(k)})} \right\rangle \quad e^{-\beta A_0} = \int d^N \vec{r} e^{-\beta U(\vec{r})}$$

$$P_k(q) = e^{-\beta(A_k - A_0)} \beta W_k(q, s^{(k)}) \tilde{P}(q, s^{(k)})$$

$$P(q) = \sum_{k=1}^n C_k(q) P_k(q) \quad \sum_{k=1}^n C_k(q) = 1 \quad \tilde{P}(q, s^{(k)}) \sim \frac{1}{n_k \Delta q} \tilde{H}_k(q)$$

$$\hat{\sigma}_k^2 = \frac{c_k(q) \tilde{H}_k(q)}{n_k \Delta q} \quad \sigma_k^2 = e^{-2\beta(A_k - A_0)} e^{2\beta W_k(q, s^{(k)})} \hat{\sigma}_k^2 \quad \sigma^2 = \sum_{k=1}^n C_k^2(q) \sigma_k^2$$

$$C_k(q) = \frac{n_k e^{\beta A_k} e^{-\beta W_k(q, s^{(k)})}}{\sum_{j=1}^n n_j e^{\beta A_k} e^{-\beta W_k(q, s^{(j)})}}$$

$$P(q) = \frac{\sum_{k=1}^n n_k \tilde{P}(q, s^{(k)})}{\sum_{k=1}^n n_k e^{\beta(A_k - A_0)} e^{-\beta W_k(q, s^{(k)})}} \quad e^{\beta(A_k - A_0)} = \int dq P(q) e^{-\beta W_k(q, s^{(k)})}$$

Rare events (contd)

Wang-Landau sampling

$$A(\vec{r}_2|\vec{r}_1) = \min \left[1, \frac{\Omega(E_1)}{\Omega(E_2)} \right] = \min \left[1, \frac{e^{-\beta U(\vec{r}_2)} e^{-h(s_1)}}{e^{-\beta U(\vec{r}_1)} e^{-h(s_2)}} \right]$$

$$h(s_2) \rightarrow h(s_2) + \alpha \Leftrightarrow g(s_2) \rightarrow fg(s_2) \quad \alpha = \ln f$$

Advanced methods

Adiabatic dynamics

$$Q(N, V, T) = C_N \int d^N \vec{p} d^{3N} q e^{-\beta \left[\sum_{i=1}^N \frac{p_i^2}{2m_i} + \tilde{V}(q_1, \dots, q_{3N}, \beta) \right]}$$

$$\tilde{V}(q_1, \dots, q_{3N}, \beta) = \tilde{U}(q_1, \dots, q_{3N}) - kT \ln J(q_1, \dots, q_{3N})$$

$$\tilde{H}(q, p, \beta) = \sum_{\alpha=1}^{3N} \frac{p_{\alpha}^2}{2m_{\alpha}} + \tilde{V}(q_1, \dots, q_{3N}, \beta)$$

TAMD: $\dot{q}_{\alpha} = \frac{p_{\alpha}}{m_{\alpha}} \quad \dot{p}_{\alpha} = -\frac{\partial \tilde{V}}{\partial q_{\alpha}} - \frac{p_{\alpha}}{Q_1} p_{\alpha} \quad \dot{p}_{\alpha} = -\frac{\partial \tilde{V}}{\partial q_{\alpha}} - \frac{p_{\alpha}}{Q_2} p_{\alpha} \quad \dot{\eta}_j = \frac{p_{\eta_j}}{Q_j}$

$$\dot{p}_{\eta_1} = \sum_{\alpha=1}^n \frac{p_{\alpha}^2}{2m_{\alpha}} - nkT_q \quad \dot{p}_{\eta_2} = \sum_{\alpha=n+1}^{3N} \frac{p_{\alpha}^2}{2m_{\alpha}} - (3N - n)kT$$

$$A(q_1, \dots, q_n) = -kT_q \ln P_{adb}(q_1, \dots, q_n) + const$$

Metadynamics

$$P(s_1, \dots, s_n) = \left\langle \prod_{\alpha=1}^n \delta(f_{\alpha}(\vec{r}_1, \dots, \vec{r}_N) - s_{\alpha}) \right\rangle$$

$$P(s_1, \dots, s_n) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T dt \prod_{\alpha=1}^n \delta(f_{\alpha}(\vec{r}_1, \dots, \vec{r}_N) - s_{\alpha})$$

$$P(s_1, \dots, s_n) = \lim_{T \rightarrow \infty} \lim_{\Delta s \rightarrow 0} \frac{1}{T} \frac{1}{\sqrt{2\pi \Delta s^2}} \int_0^T dt \prod_{\alpha=1}^n e^{-\frac{(s_{\alpha} - f_{\alpha}(\vec{r}_1(t), \dots, \vec{r}_N(t)))^2}{2\Delta s^2}}$$

$$U_G(\vec{r}_1, \dots, \vec{r}_N, t) = W \sum_{t-T_G, 2T_G, \dots} e^{-\sum_{\alpha=1}^n \frac{(f_{\alpha}(t) - f_{\alpha}(t_{G(t)}))^2}{2\Delta s^2}}$$

Transition path ensemble

$$x_{(k+1)\Delta t} = e^{iL_2 \frac{\Delta t}{\hbar}} e^{iL_1 \Delta t} e^{iL_2 \frac{\Delta t}{\hbar}} \equiv \phi_{\Delta t}(x_{k\Delta t})$$

$$T(x_{(k+1)\Delta t} | x_{k\Delta t}) = \delta(x_{(k+1)\Delta t} - \phi_{\Delta t}(x_{k\Delta t}))$$

$$\mathcal{P}[\mathcal{X}(T)] = f(x_0) \prod_{k=0}^{n-1} T(x_{(k+1)\Delta t} | x_{k\Delta t}) \quad f(x_0) = \frac{e^{-\beta \mathcal{H}(x_0)}}{Q(N, V, T)}$$

$$\mathcal{P}_{AB}[\mathcal{X}(T)] = \frac{1}{\mathcal{F}_{AB}(T)} h_A(x_0) \mathcal{P}[\mathcal{X}(T)] h_B(x_{n\Delta t})$$

$$\mathcal{F}_{AB}(T) = \int dx_0 \dots dx_{n\Delta t} h_A(x_0) \mathcal{P}[\mathcal{X}(T)] h_B(x_{n\Delta t}) = \int dx_0 h_A(x_0) f(x_0) h_B(x_{n\Delta t}(x_0))$$

$$\mathcal{R}_{AB}[X(T)|Y(T)] = \Lambda_{AB}[X(T)|Y(T)] \mathcal{T}_{AB}[X(T)|Y(T)]$$

$$\Lambda_{AB}[X(T)|Y(T)] = h_A(x_0) h_B(x_{n\Delta t}) \min \left[1, \frac{f(x_0)}{f(y_0)} \right]$$

The committor distribution

$$P(p) = \frac{C_N}{Q(N, V, T)} \int d^N \vec{p} \int_{q(\vec{r})=q^{\dagger}} d^N \vec{r} e^{-\beta \mathcal{H}(\vec{r}, \vec{p})} \delta(p_B(\vec{r}_1, \dots, \vec{r}_N) - p)$$

Physical chemistry

Principles

Conversion Celsius to Kelvin: $\theta/C = T/K - 273.15$

Additions

Quantum mechanics

Energy levels of a molecule with mass m in a container of length L: $E_n = \frac{n^2 h^2}{8mL^2}$ with $n = 1, 2, 3, 4, \dots$

Mathematics

Binomial coeff when N to infinity: $\frac{N!}{n_1! n_2! \dots n_n!} = \frac{N^N}{n_1^{n_1} n_2^{n_2} \dots n_n^{n_n}}$

Saddle point: when $H = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}(x_0, y_0)^2 \neq 0$.

Variance: $\langle (\Delta A)^2 \rangle = \langle (A - \langle A \rangle)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2$

Geometric series and sequence

Geometric sequence is collection of terms, series is the sum.

Nth value: $a_n = a_1(r)^{n-1}$ where r is the common ratio

Partial sum: $S_n = \frac{a_1[1-r^n]}{1-r}$

The geometric mean $\sqrt{n_a n_b}$ gives the value in the middle between a and b of the sequence

Sum of infinite series: $S_{\infty} = \frac{a_1}{1-r}$ if series converges ($|r| < 1$), otherwise diverges

Thermodynamic relationships

First law: $\Delta U = \Delta Q - \Delta W$

Second law: $dS = \frac{dQ_{rev}}{T}$

Third law: $S(T = 0K) = 0$ Therefore $dU = TdS - PdV + \mu dN$

Specific heat (Heat capacity)

$$C_V = \left(\frac{\partial E}{\partial T} \right)_{V, N} = T \left(\frac{\partial S}{\partial T} \right)_{V, N}$$

$$C_P = \left(\frac{\partial H}{\partial T} \right)_{P, N} = T \left(\frac{\partial S}{\partial T} \right)_{P, N}$$

Thermal expansion: $\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_{P, N}$

Isotherm. compressibility: $\alpha = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$

Adiabatic compressibility: $\alpha = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_{S, N}$

$k = 1.38 * 10^{-23} JK^{-1}$

When quantities x, y, z are mutually related

$$\left(\frac{\partial x}{\partial y} \right) = 1 / \left(\frac{\partial y}{\partial x} \right)$$

$$\left(\frac{\partial}{\partial z} \left(\frac{\partial y}{\partial x} \right) \right)_z = \left(\frac{\partial}{\partial x} \left(\frac{\partial y}{\partial z} \right) \right)_x$$

$$\left(\frac{\partial x}{\partial y} \right)_z \left(\frac{\partial y}{\partial z} \right)_x \left(\frac{\partial z}{\partial x} \right)_y = -1$$

$$\left(\frac{\partial x}{\partial y} \right)_w = \left(\frac{\partial x}{\partial y} \right)_z + \left(\frac{\partial x}{\partial z} \right)_y \left(\frac{\partial z}{\partial y} \right)_w$$

Thermodynamic relationships (Atkins)

$$U - U(0) = - \left(\frac{\partial \ln Q}{\partial \beta} \right)_V \quad S = \frac{U - U(0)}{T} + k \ln Q \quad p = kT \left(\frac{\partial \ln Q}{\partial V} \right)_V$$

$$H - H(0) = - \left(\frac{\partial \ln Q}{\partial \beta} \right)_V + kTV \left(\frac{\partial \ln Q}{\partial V} \right)_T \quad \text{when ideal gas: } H - H(0) = \frac{5}{2} nRT$$

$$G - G(0) = -kT \ln Q + kTV \left(\frac{\partial \ln Q}{\partial V} \right)_T$$

when ideal gas $G - G(0) = -kT \ln Q + nRT = -nRT \ln \frac{q_m}{N_A}$

The energy can be given from a series of contributions: $\epsilon_i = \epsilon_i^T + \epsilon_i^R + \epsilon_i^V + \epsilon_i^E$ (Translation, Rotation, Vibration, Electronic) $\rightarrow q = q^T q^R q^V q^E$

$$q^T = V/\Lambda^3 \quad \Lambda = h \left(\frac{\beta}{2\pi m} \right)^{1/2}$$

$$q^R = \sum_J (2J + 1) e^{-\beta h c B J(J+1)}$$